

SVM training and prediction (non-linear)

Use the dataset, train a Gaussian kernel SVM, called 'SVMModel_nonlinear' on the training data.

Use it to predict the labels of the test data and compute the accuracy percentage.

Script

 Save

 Reset

 MATLAB Documentation (https://www.mathworks.com/help/)

Open Item in MATLAB Online 

```
1 % Load data
2 data = readtable('fruit_dataset.csv');
3 X = data(:, 2:6);
4 Y = data(:, 'name');
5 % Split into training and test data by the ratio 75% and 25%, respectively.
6 cvObj = cvpartition(size(X, 1), 'HoldOut', 0.75);
7 idx = cvObj.test;
8 X_train = X(idx, :); X_test = X(~idx, :);
9 Y_train = Y(idx, :); Y_test = Y(~idx, :);
10
11 % Train a nonlinear SVM with Gaussian kernel. Should the data be standardized? Your code:
12 SVMModel_nonlinear = fitcsvm(X_train, Y_train, 'KernelFunction', 'gaussian', 'Standardize', true);
13 % Predict using the nonlinear SM, your code:
14 Y_pred_nonlinear = predict(SVMModel_nonlinear, X_test);
15 accuracy_nonlinear = sum(Y_pred_nonlinear == Y_test) / length(Y_test) * 100;
```

 Run Script 

Tests: All Tests Passed

Submit 

-  Train a nonliear SVM model that can judge whether a given fruit is orange or grapefruit.
-  Predict the nonlinear SVM on the test dataset.
-  Calculate the prediction accuracy of the nonlinear SVM model. The variable of the prediction accuracy must be named to 'accuracy_nonlinear'.

Output

Code ran without output.